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Beech regeneration research: From ecological to silvicultural aspects

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ABSTRACT

This review describes key regeneration characteristics of the genus *Fagus* as represented by its four most prominent species: *F. crenata* (*F.c.*), *F. grandifolia* (*F.g.*), *F. orientalis* (*F.o.*) and *F. sylvatica* (*F.s.*). Similarities and differences in the relevant life phases of these species are identified. Those are related to natural disturbance regimes and synecological peculiarities of the forests where they grow, thereby establishing a basis for evaluating the likely outcome of different silvicultural measures.

Common ecological characteristics of these *Fagus* species' life cycles include the masting phenomenon, pollen dispersal with effective distances of about 100 m, seed dispersal to about 20 m, seedling sensitivity to frost, drought, and animal predation, and a very shade tolerant establishment phase. This commonality suggests its appropriateness as a "model-genus". However, some species also have unique ecological characteristics not observed in the others. *F.g.* exhibits root suckering, and beech bark disease seems to trigger vegetative regeneration by that means. Likewise, its masting behaviour deviates from *F.s. F.o.* and *F.c.*, occurring more frequently and more regularly. In *F.c.* forests, dwarf bamboo species and their ecological characteristics are important determinants of tree regeneration establishment.

The small canopy gaps that commonly occur in *Fagus* dominated natural forests fit very well with the genus' regeneration characteristics. These conditions are best duplicated by management measures, which maintain partial overstory shading until the seedlings are large enough for release. However, such a strategy reduces chances to regenerate more light-demanding associated species. Together with differences in landowner objectives, the diversity of ecological conditions within and between the species of *Fagus* requires site-specific prescriptions to insure regeneration success, e.g. cutting regimes.

Of particular interest to research are the challenges of managing mixed-species stands for high quality timber production in Central European and Caspian beech forests, the decline of *F.g.* and how to deal with the aftermath forest, and effective ways to manage *F.c.* in coexistence with dwarf bamboo. Further, the historic dispersal of heavy seeded *Fagus* species over long distances is still poorly understood. In addition, since their drought sensitive seedlings may be damaged or killed during extreme weather, research must address the possible effects of global climate change on the regeneration potential of beech forests. Species-bridging research may be needed to address these questions.

1. Introduction

Pressure to apply ecologically sound practices and make decisions more transparent requires forest scientists to explore new pathways for managing forest stands. Global dialogue provides ideas and solutions. But silviculture must ultimately account for regional differences (e.g., species characteristics and mixtures, environment, and management objectives) that affect the outcome of alternative treatments. As a contribution, we introduce the concept of a "model genus" using the species of *Fagus* as an example. A similarity in their features allows the integration of information from multiple, independent studies, and provides a basis for evaluating alternative management strategies for beech dominated forests.

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Among features that make *Fagus* suitable to this approach are:

- its distribution throughout the temperate northern hemisphere (Peters, 1997), and the range of associated environmental conditions;
- the ecological similarity of its species as representatives of ones having shade-tolerant, dominant climax traits;
- the amount and quality of available knowledge about its biology and management; and
- the economical importance of its component species.

This review makes that vision real by synthesizing detailed aut- and synecology research findings about *Fagus* regeneration, and articulating the similarities and differences in ecologic characteristics and silviculture for the genus around the world. It primarily compares *Fagus crenata* Blume (*F.c.*), *Fagus grandifolia* Ehrh. (*F.g.*), *Fragus orientalis* Lipsky (*F.o.*) and *Fagus sylvatica* L. (*F.s.*). Oriental beech *(F.o.)* is included as a species, as it has its own morphological trait. We are aware of the ongoing discussion about the species status of *F.o.* as it has few unique alleles (see Denk et al., 2002; Salehi Shanjani and Sagheb-Talebi, 2006; Paffetti et al., 2007). Only limited information about other *Fagus* species is available, which were not taken into account in the review.

We structured the review around the different life cycle phases of these species (after Harper, 1977), including:

- flowering and fruiting phase
- seed dispersal phase
- establishment phase

By comparing both differences and similarities among the species of *Fagus*, we identified the risks of failure in management measures and the means to overcome them. Such an approach insures that silviculture accounts for the critical ecologic factors and results in an acceptable vitality, density, survival, growth, and quality of regeneration at an appropriate time.

The review: (i) tests the applicability of the model genus idea using *Fagus*; (ii) identifies regeneration problems with different *Fagus* species; (iii) discusses options for reducing the risks; and (iv) identifies research needs.

2. Beech regeneration in the context of forest dynamics

2.1. Pattern of forest disturbance, natural regeneration and the silvicultural implications

The majority of North American and European beech forests have been logged or once cleared for agricultural use (Peters, 1997), followed by replenishment or recolonization through natural regeneration. In fact, primary forests having an important component of beech are absent or extremely rare on these two continents (Cronon, 1983; Bradshaw, "Holocene history of beech forests", this volume). However, Japan still has about 1.4 million ha of primary forest (Ministry of Environment, 1997) and Iran more than 100,000 ha with *F.o.*as an important species (Sagheb-Talebi et al., 2004; Knapp, 2005). These remaining areas provide a valuable reference for studying temperate deciduous forest ecosystem dynamics, and when considering the silvicultural options for beech forests in general.

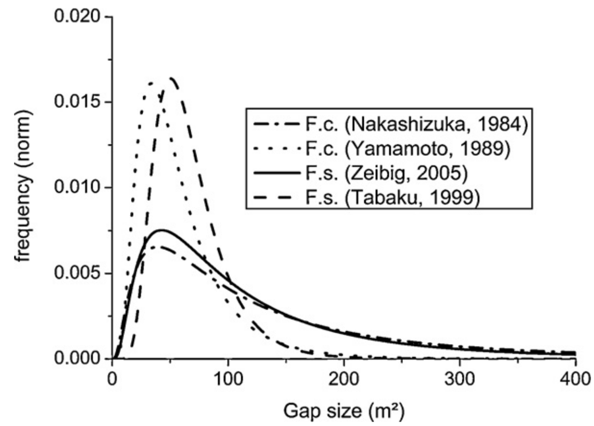

Fig. 1. Canopy gap size frequency distributions of four beech forest reserves. Log-normal distributions were computed for two *F.c.* forests (data published in Nakashizuka, 1984; Yamamoto, 1989) and two *F.s.* forests (data published in Tabaku and Meyer, 1999; Zeibig et al., 2005) by maximum-likelihood method.

Forest disturbance is central to regeneration responses of *Fagus* throughout the globe, both in primary forests and replenished or managed ones. In that sense, "disturbance" includes anything that initiates regeneration from seeds, affects the growth dynamics of advance seedlings and saplings, or induces root suckering or stump sprouting.

2.1.1. Pattern of forest disturbance

Disturbance in beech forests may range from large-scale stand-replacing events (blowdown during a storm), to large openings once occupied by a group of trees, to small single tree gaps. Mortality of small or intermediate trees due to inter-tree competition also open limited-size gaps in the canopy (Nakashizuka et al., 1992; Delfan Abazari et al., 2004a,b). These disturbances may alter the forest structure without reducing canopy closure, as with the dominance and synchronous death of dwarf bamboos (*Sasa* spp.) in *F.c.* ecosystems (Abe et al., 2001), or when disturbance to shallow roots triggers root suckering of *F.g.* (Nyland et al., 2006a). Prolonged browsing during times of high deer densities has also altered the composition and abundance of tree seedlings and other vegetation in *F.s.* and *F.g.* stands (e.g. Sage et al., 2003). Air pollution, nutrient deposition, and altered ground water tables may also affect forest structure and regeneration. Even so, "disturbance" in temperate regions is commonly equated with openings (gaps) in the overstory canopy. Further, the multi-age and irregular structure

Table 1
Indicating variables of canopy gap size frequency distributions of some beech forest reserves. Log-normal distributions were computed by maximum-likelihood method. Values of indicating variables were taken from those computations.

Species, authors, name of reserve, area considered	Average gap size (m^2)	Mode of gap size (m^2)	Proportion of total gap area in gaps smaller than 200 m^2	Proportion of reserve area under gaps, gap definition
F. crenata, Nakashizuka, 1984, Mt. Moriyoshi, 2.4 ha	161	39	0.40	0.20 canopy gaps
F. crenata, Yamamoto, 1989 Miscellaneous, 20.0 ha	59	34	0.95	0.12 canopy gaps
F. sylvatica, Zeibig et al., 2005, Krokar, 12.0 ha	129	43	0.53	0.056 canopy gaps
F. sylvatica, Tabaku and Meyer, 1999, Mirdita, 5.0 ha	74	39	0.88	0.066 canopy gaps
F. sylvatica, Tabaku and Meyer, 1999, Puka, 3.6 ha	62	46	0.99	0.034 canopy gaps
F. sylvatica, Tabaku and Meyer, 1999, Rajka, 6.0 ha	67	50	0.99	0.033 canopy gaps

of primary *F.o.* stands of northern Iran (Eslami and Sagheb-Talebi, 2007; Shahnavazi et al., 2005; Sagheb-Talebi and Schütz, 2002) suggests that the patterns of disturbance and regeneration may differ among developmental stages and stand types within a region.

2.1.1.1. Gap size distribution. Canopy disturbance promotes the growth of beech regeneration into upper canopy positions. Thus, the frequency and sizes of gaps profoundly affects the dynamics of beech communities. In turn, the species composition of a stand and local environmental conditions temper the disturbance regime. However, within pure beech forests, gaps are primarily small (*F.c.*: Nakashizuka, 1987; Yamamoto, 1989; *F.g.*: Runkle, 1982; *F.s.*: Butler Manning, 2007).

Fig. 1 and Table 1 describe gap frequencies for different forests having pure beech stands. They show that gaps smaller than 200 m^2 are frequent in old-growth beech-dominated systems (mean gap size of 59 to 161 m^2). With *F.c.* and *F.s.* most gaps are between 34 and 50 m^2, similar to the crown size of old beech trees and suggesting a single tree mortality pattern of gap formation. While naturally created forest gaps in most beech forests may cover between 3 and 20% of the area (Table 1), Japanese forests have 12–20% in gaps (Nakashizuka, 1987).

2.1.1.2. Disturbance caused by beech bark disease. Understory *F.g.* density has increased as beech bark disease spread across northeastern North America following the introduction of *Cryptococcus fagisuga* (Lindinger) from Europe. The scale insect creates openings in the bark. These serve as infection courts for a *Nectria* fungus that kill patches of cambium, eventually girdling the tree (Houston, 2005). Since only about 1% of all *F.g.* may have a resistance, tolerance, or immunity (Houston, 1997), few trees larger than 20–25 cm diameter escape its effect (Houston, 1997). Similar symptoms have recently been reported in Caspian beech forests (Kiadaliri et al., 2008).

Where *F.g.* has been abundant in a stand, beech bark disease opened the overstory canopy appreciably, and root suckers have become dense throughout the understory (Nyland et al., 2006a). Within stands having only scattered *F.g.* trees, the mortality creates dispersed small gaps, and root suckers mostly encircle the dying trees.

2.1.1.3. Disturbance within stands having a dwarf bamboo understory. Survival and development of *F.c.* regeneration depends on light availability, and this is influenced by the amount of dwarf bamboo biomass in an understory (Maeda, 1988; Abe et al., 2001, 2002). The dwarf bamboo flowers once in several decades and then dies over even several square kilometres, dramatically changing the understory light regime (Nakashizuka, 1987; Abe et al., 2001) and stimulating growth of any advance beech regeneration. Hence, synchronous canopy gap formation and bamboo dieback seem necessary for beech regeneration to become established and grow.

2.1.2. Sexual regeneration

2.1.2.1. Flowering, pollen dispersal, and fruiting. Male and female flowers of beech occur separately on the same tree, and are vulnerable to spring frosts (Savill, 1991; Young and Young, 1992). *F.s.* pollen effectively disperses less than 250 m within forests (Wang, 2001; Gerber in Kramer, 2004), and *F.c.* up to 193 m (Shimatani et al., 2007). Pollen may disperse for 500 m in some cases, but generally only up to 60 m in high density stands (Shimatani et al., 2007).

Evidence relates seed production to tree diameter (for *F.s.*: Wagner, 1999), with flowering and seed production beginning at about age of 40 to 50 in *F.g.*, *F.c.*, and *F.s.* Major masting events in the later two species occur at 2- to 20-year intervals (Watt, 1923; Young and Young, 1992; Kon et al., 2005). *F.o.* generally begins fruiting at age 60 with masting every 3 to 18 years (Mirbadin and Gorji, 1996; Mirbadin and Namiranian, 2005). By contrast, intervals of 2-3 years seem common for *F.g.* (Schopmeyer, 1974; Jacabus et al., 2005; McNulty and Masters, 2005). And while seed production among trees >25–30 cm dbh decreased by two-thirds to three-fifths as beech bark disease progressed (Costello, 1992), masting recently increased due to modest seed production on large numbers of smaller trees that eventually reached pole stage (McNulty and Masters, 2005).

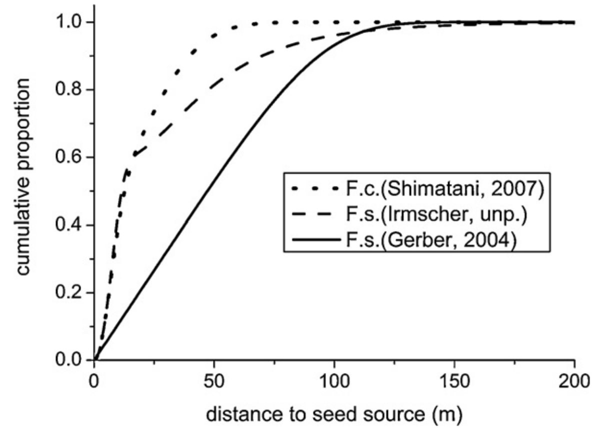

Fig. 2. Cumulative beech dispersal models derived from established seedlings distributions. Models of Shimatani et al. (2007) for *F.c.*, and Gerber (in Kramer, 2004) for *F.s.* based on genetic analysis in pure stands. Model by Irmscher (unpubl. data) for *F.s.* based on the distribution of established *F.s.* seedlings from a single mother tree in a pure spruce (*Picea abies*) stand.

Good seed production in *F.s.* and *F.g.* depends on climatic events during two consecutive years having conditions that favour carbohydrate build-up, followed by an early summer drought the next year (Piovesan and Adams, 2001). A warm July portends good flowering of *F.s.* the following year (Övergaard et al., 2007), and frequency of masting has positively correlated with site index. *F.o.* flowering and fruiting also depend on climate and site conditions, with heavy seed production correlated with soil nutrition, northern aspects, and an altitude of around 1500 m.a.s.l. Seeds also had greater size and weight in stands at 750 to 1500 m.a.s.l. (Etemad and Marvi Mohajer, 2004). Fruiting of *F.o.* in the Caspian region varies from tree to tree, and site to site. That, along with the frequent masting of *F.g.*, seems contradictory to the hypothesis for a climatic linkage to beech masting, and consistent with observations by Masaki et al. (2008) that masting events in *F.c.* differ among the districts in northern Japan.

Theory predicts an evolutionary benefit for masting, particularly with respect to pre-dispersal seed predation (Sork, 1993; Yasaka et al., 2003; Piovesan and Adams, 2005). Different factors seem to control these events with *F.s.* and *F.c.*, compared to *F.g.* (Piovesan and Adams, 2001), but they are poorly understood. This problem is much more pressing in Japan, where managers need to predict the timing of good seed years for *F.c.* so they can schedule advance clearing of dwarf bamboo from the understory. Yet after considering several hypotheses forwarded by Kelly and Sork (2002) and Kelly (2004), Yasaka et al. (2003) and Kon et al. (2005) concluded that escape from predation and pollination efficiency are most important determinants for *F.c.*

2.1.2.2. Dispersal. Beechnuts commonly disperse by barochory, usually for up to 20 m. Yet more than 80 m has been observed with *F.c.* and *F.s.* Models of established beech regeneration for *F.c.* by Shimatani et al. (2007) and *F.s.* by Gerber (in Kramer, 2004) appear in Fig. 2, along with one by Irmscher (unpubl. results). They suggest distances of up to 125 m, with zoochorous dispersal even introducing beech into stands of other species.

Small mammals generally move nuts of *F.c.* and *F.g.*, but for relatively short distances (Tubbs and Houston, 1990; Young and Young, 1992; Miguchi, 1996). Yet Kunstler et al. (2004) found *F.s.* natural regeneration in pine stands up to 3,000 m from nearest beech trees, compared to less than 300 m into open grassland. Jays seemed primarily responsible for the longer dispersal. The birds also may carry *F.g.* seeds for several kilometres (Johnson and Adkisson, 1985; Tubbs and Houston, 1990).

2.1.2.3. Seed wintering and germination. In *F.g.,* burs open in autumn after freezing temperatures, commonly releasing two nuts per bur (Burns and Honkala, 1990). The seeds fall to the forest floor in a dormant state and remain so throughout autumn and winter. *F.s.* nuts need several months of this pre-chilling, depending on the temperatures during that period (Tubbs and Houston, 1990; Gosling, 1991; Young and Young, 1992). Individual seeds and different seedlots vary in these requirements, but natural stratification in a moist seedbed usually pre-conditions them. Since germination and sprouting are temperature dependent (Harper, 1977), the time of germination varies with spring weather conditions (Runkle, 1989). Radicles may emerge several months before the embryonic shoot develops into a recognizable beech seedling (Madsen, 1995; Farmer, 1997; Young and Young, 1992).

The requisite pre-chilling complicates programs of artificial regeneration with *Fagus* (i.e., nursery production or direct seeding), since managers depend on reliable and predictable germination following sowing (Willoughby et al., 2004a; Baumhauer et al., 2005). Bonner and Leak (2008) described techniques for pre-chilling an entire beech seedlot at 28–30% moisture content until dormancy is broken in most of the seed, then increasing moisture to initiate uniform germination.

Beechnuts landing on natural seedbeds are susceptible to unfavourable environmental conditions (e.g. desiccation and frost), harmful fungi, and predation (insects, birds, and both large and small mammals). The most important predators of *F.s.* are rodents such as *Apodemus sylvatica*, *Clethrionomys glareolus*, birds and wild boar (*Sus scrofa*) (Burschel et al., 1964; Harmer, 1995; Ammer et al., 2002; Willoughby et al., 2004b). Survival has been improved dramatically by seedbed preparation shortly before, during, or after natural seed fall or direct seeding (See Burschel et al., 1964; Huss and Burschel, 1972; Huss and Stephani, 1978; Jungbluth and Dimitri, 1980; Madsen, 1995). For direct seeding, Ammer et al. (2002) demonstrated that mineral soil seedbeds provide a stable moisture regime and soil temperatures favourable to beechnut germination. In fact, first-year regeneration density may be 100-fold greater in prepared seedbeds than on the untreated forest floor (Olesen and Madsen, 2008). Liming acidic soil also improved seedling density and growth following direct seeding (Küssner and Wickel, 1998; Ammer et al., 2002), but had no effect at less acidic sites.

Studies of mixed soil and mineral seedbeds did not reveal whether lower regeneration density in the former may result from increased fungal attack due to the higher organic matter content in the soil, or because the mixed seedbed facilitated rodent tunnelling and seed predation (Burschel et al., 1964; Dubbel, 1989; Madsen, 1995). Snow cover has reduced predation of *F.c.* nuts (Shimano and Masuzawa, 1998; Homma et al., 1999).

Many birds, mammals, and insects eat or damage *F.g.* and *F.s.* seeds, and they affect dispersal by collecting and storing beechnuts in caches. Beechnut availability may affect the survival and reproduction of these creatures during harsh winters (Jensen, 1985; Yasaka et al., 2003; Jacabus et al., 2005), with rodent populations increasing following major beech masting events (Jensen, 1982). Calculations of potential consumption based on energy requirements suggest that small animals may consume significant proportions of the beechnut crop during years of limited masting. However, during major masts years the loss to all kinds of predation will have little effect on overall seed supply (Jensen, 1982; Yasaka et al., 2003; Kon et al., 2005; Olesen and Madsen, 2008).

2.1.2.4. Establishment and early growth.

2.1.2.4.1. Effects of different canopy openings. Beech seedlings become established under a wide range of canopy openings (Sagheb-Talebi and Schütz, 2002). They survive for long periods at very low light levels (Relative Light Intensity, RLI = 1%) (Emborg, 1998; Modry et al., 2004), but grow slowly (Gansert and Sprick, 1998; Collet et al., 2001). Height and diameter growth are best in the open (RLI = 100%), but differ little with light at 30% < RLI < 50% (Gemmel et al., 1996; Kunstler et al., 2005). Beech seedlings often undergo multiple suppression-and-release episodes before reaching the upper canopy (Nagel et al., 2006; Collet et al., 2008). Even after a long period of suppression, height growth increases following each canopy disturbance, (Nakashizuka, 1983; Canham, 1990; Collet and Chénost, 2006) and particularly after the second and third release (Leak, 2003).

Seedlings have a highly plastic morphology that depends on genetics, light, water, nutrient availability, and frost occurrence (Thiébaut et al., 1985; Nicolini, 1997). Beech is characterized by a monopodial branching pattern, a plagiotropic trunk secondarily reoriented into a vertical position by cambial activity, and plagiotropic branches (Peters, 1997; Hallé et al., 1978). Shoot elongation is rhythmic, with two growth flushes per season at good sites and often sympodial due to frequent death of the apical bud (Sagheb-Talebi, 1996; Roloff, 1986). At RLI > 40%, all main axes orient into a vertical position, but multiple growth flushes and the rapid growth induced by high light result in large branches and forks. With RLI < 10%, the branches and main axis of beech do not reach a vertical position due to the low radial growth (Nicolini et al., 2001), negatively affecting sapling architecture (for *F.s.*: Diaci and Kozjek, 2005; for *F.g.*: Canham, 1988; Nyland et al., 2006a). However, branches and forks remain small in diameter and do not compete strongly with the main axis (Bonosi, 2006). At light 20% < RLI < 40%, *F.s.* and *F.o.* seedlings have a better morphology, with weaker lateral branches and forks than at high light levels and greater verticality of the main axis than at low light levels (Sagheb-Talebi et al., 2001).

Overtopping mature trees and neighbouring understory vegetation may shade beech seedlings. Yet moderate to dense shading over long periods may result in better-formed beech saplings (Leonhardt and Wagner, 2006). At full light, beech will have good morphology only when grown at a high seedling density of conspecifics (Sagheb-Talebi and Schütz, 2006) or with neighboring vegetation of comparable density. To that end, recommendations for low-density plantations (e.g. fewer than 1500 tree per ha) include maintaining or establishing neighbouring woody vegetation to provide the necessary lateral shade.

2.1.2.4.2. Limitations in analysis of single factors. Most studies of beech seedling responses have not separated effects of competition for various resources, complicating any extrapolation to sites with different water or nutrient availabilities (but see Wagner et al., 2009). Height growth at high light is much less at low than at high water availability (Vinkler et al., 2007) and growth responses after changes in light availability also depend on water availability (Madsen, 1994). Seedling growth has also been related to light availability and root density of old beech (Wagner, 1999), with *F.s.* seedlings recovering more quickly in good light after trenching excluded roots of older trees (Fig. 3). Overall, RLI seems the best indicator of environmental change, with morphology of young beech differing under varying light environments.

2.1.2.4.3. Ground vegetation as competitor. Dwarf bamboo prevents development of even advance *F.c.* regeneration (Abe et al., 2002), and associated ground vegetation affects *F.s.* seedling growth

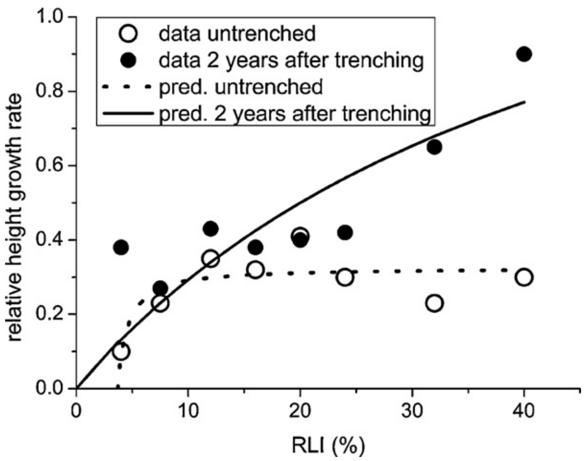

Fig. 3. *F.s.* seedling growth related to Relative Light Intensity (RLI in percent of that above canopy). Open circles indicate relative height growth prior to trenching in a pure beech stand. Solid circles indicate relative height growth in the same plots 2 years after root trenching. The predictions are based on the Michaelis-Menten function $\mathrm{RHG} = A\left[(\mathrm{RLI}-B)/\left(\left(A/C\right)+(\mathrm{RLI}-B)\right)\right]$, with RHG as relative height growth, RLI as relative light intensity and A, B and C the parameters. Data from Wagner (1999).

(e.g., Burschel and Schmaltz, 1965; von Lüpke, 1987). While beech seedlings and saplings withstand interference by most weeds (Madsen, 1995), they must be weeded to insure satisfactory height growth when planted or sowed on former farmland (Löf, 2000; Löf et al., 2004). Much depends on the density of ground vegetation and the effect of any overstory trees.

Most ground vegetation does not thrive as well as young beech beneath a moderately closed *F.s.* canopy, lessening its competitive effects. In fact, site preparation using herbicides did not improve *F.s.* seedlings biomass production when applied under a dense shelter, compared to a light overhead cover (Huss and Stephani, 1978), suggesting use of moderate canopy openings to establish *F.s.* regeneration where competition from ground vegetation might be strong (Kühne and Bartsch, 2003).

Compared to oaks, planted *F.s.* seedlings are more prone to mice and vole damage in areas with weed cover (von Lüpke, 1987). With *F.c.*, predation by rodents beneath closed-canopy stands decreases following the dieback of dwarf bamboo, resulting in increased seed and seedling survival (Abe et al., 2005).

2.1.3. Vegetative regeneration

2.1.3.1. Sprouting and root suckers. While *F.c.*, *F.g.*, *F.o.* and *F.s.* sprout from cut stumps, coppice systems have been developed only for *F.c.* and *F.s.* (Peters, 1997). In *F.g.*, stump sprouts originate either from adventitious buds around the top of a stump, or from dormant buds along the sides (Mallett, 2002; Nyland et al., 2006a). Most die within 2–3 years (Mallett, 2002), but some persist off stumps in the open or beneath a low-density overstory (Nyland, unpublished results). Sprouting potential decreases after *F.g.* and *F.s.* trees reach 10 cm dbh, and most *F.g.* stump sprouts do not develop into trees (Eyre and Zillgitt, 1953; Hamilton, 1955; Fowells, 1965).

F.g. produces root suckers. These may form a dense understory following canopy disturbances (Houston, 1975; Krasny and DiGregorio, 2001; Nyland et al., 2006a; Runkle, 2007), usually within 9–10 m around the source trees (Jones and Raynal, 1987; Nyland et al., 2006a). Most arise from wounds to shallow and exposed roots (Jones and Raynal, 1987), and often following logging (Houston, 1975; Jones and Raynal, 1987; Houston, 2001). Yet dense root suckers have developed even beneath unmanaged stands (Nyland et al., 2006a). As the suckers develop, new ones originate off their extending roots, so that groups of overstory *F.g.* trees may originate from a common source (Houston and Houston, 1987). As a result, a stand with many individual *F.g.* stems may have relatively few separate clones.

Among advance *F.g.* regeneration <1.2 m tall, seedlings may be more abundant than suckers at some sites, but not at others (Nyland, 2008). Therefore, each *F.g.* clone may have a characteristic flowering and suckering tendency. Even so, at sites having many small *F.g.* seedlings, root suckers commonly dominate the advance beech >0.6 m tall (Nyland, 2008).

Root suckers have become dense in unmanaged stands where beech bark disease weakened or killed the large *F.g.* (Nyland et al., 2006b), including unmanaged old-growth communities (McNulty and Masters, 2005). This implicates the disease as a probable contributing agent. Surface disturbance and light near the ground seem linked to root suckering in managed stands infected with beech bark disease (Nyland et al., 2006a). Yet root suckers may be more abundant around resistant beech trees and their stumps, than around diseased trees and their stumps (Houston, 2001).

2.2. Competition and predation in the establishment phase—impact to mixtures

2.2.1. Height growth dynamics and interspecific competition

In the young stages, *F.g.* grows slower than *Acer saccharum*, *Betula alleghaniensis*, or *Fraxinus americana* (Nyland et al., 2006a). So does *F.s.* growing in mixture with *Fraxinus excelsior*, *Acer pseudoplatanus*, *Quercus petraea*, and *Q. robur* (Joyce et al., 1998; Hein et al., 2008), and *F.o.* intermixed with *Acer velutinum* (Sagheb-Talebi, 1998). Yet *Fagus* seedlings overtopped by other species generally persist for long periods, then regain and maintain dominance at later stages of stand development (for *F.c.*: Yoshida and Kamitani, 2000). Thus, it is common to grow *F.s.* in mixture with other species, waiting for it to eventually become dominant (Joyce et al., 1998). This transition usually occurs early in a rotation for mixtures of *F.s.* and small-stature pioneers (e.g., *Betula pendula*) or slowly growing species (like *Quercus petraea*, *Sorbus torminalis*), but late in a rotation for stands having high-stature post-pioneer ones (e.g., *Fraxinus excelsior*, *Acer pseudoplatanus*).

In mixed-species regeneration, the relative dominance of each varies along trophic and water gradients (Ellenberg, 1988b). On highly productive sites with rapidly growing post-pioneer species (e.g., *Fraxinus excelsior*, *Acer pseudoplatanus*), *F.s.* seedlings may succumb within a few years (Rysavy, 1991), whereas an opposite dynamics is observed on nutrient-poor sites. Similarly, with *Fagus-Quercus* mixed regeneration, *Quercus* will dominate only on drier sites (Vera, 2000).

The balance among species also depends on overstory and ground vegetation competition, herbivory, effects of pathogens, interactions with microfauna and microflora, and other abiotic factors such as late frost (Connell, 1990). Research has mainly concentrated on overstory competition and ungulate herbivory. Both strongly affect development of *Fagus* and its associated species, and both may be controlled by well designed management practices.

2.2.2. Light availability and canopy disturbance

While *Fagus* seedlings generally grow more slowly than most associated species (Beaudet and Messier, 1998; Dreyer et al., 2005), they survive better at low or intermediate light (*F.g.*, Logan, 1973; Nyland et al., 2006a,b). Thus, managers can influence the composition of mixed-species regeneration by regulating the degree of canopy cover (Abe et al., 1995; Poulson and Platt, 1996). To illustrate, with mixtures of shade-intermediate *Fraxinus excelsior* and *Acer pseudoplatanus*, maintaining a closed canopy reduces their growth compared to *F.s.* With less shade-tolerant species like *Quercus petraea*, an RLI < 30% will preferentially enhance the development of *Fagus* (von Lüpke, 1998; Shahnavazi et al., 2005; Sagheb-Talebi et al., 2001).

Height growth of *Fagus* increases after release by natural disturbance (including beech bark disease) or overstory cutting, even after long periods beneath a closed canopy (Canham, 1990; Collet et al., 2001; Nyland et al., 2006b; Nagel et al., 2006; Emborg, 2007) . Given periodic release, the beech develops into tall advance regeneration that may interfere with other species (Krasny and DiGregorio, 2001; Collet et al., 2008). In fact, stands having a dense understory of *F.g.* will lack other regeneration, necessitating beech removal to insure establishment of other species (Kelty and Nyland, 1981; Bohn and Nyland, 2003; Hane, 2003).

2.2.3. Browsing

Beech seedlings are vulnerable to birds, rodents, deer, and other herbivores (Burschel et al., 1964; Madsen, 1995; Olesen and Madsen, 2008). Deer prefer *Acer saccharum*, *Acer rubrum*, *Betula alleghaniensis*, *Prunus serotina, Fraximus Americana, Fraxinus excelsior*, *Acer pseudoplatanus*, *Carpinus betulus*, *Quercus petraea* and *Q. robur* over *F.g.* and *F.s* (Harmer, 2001; Gill, 1992; Ellenberg, 1988a; Eiberle and Bucher, 1989; Nyland et al., 2006a). They also feed on succulent *F.g.* stump sprouts and root suckers (Nyland et al., 2006a), but protracted browsing of other species commonly promotes the dominance of understory *F.g.*, even beneath uncut stands (Nyland et al., 2006a). In *F.c.* forests, browsing reduces dwarf bamboo competition, while rodent predation is heavier beneath a cover of bamboo (Wada, 1993; Abe et al., 2001). Even so, severe browsing may destroy regeneration not protected in fenced enclosures (Akashi, 1997), or unless the deer population is reduced (Akashi and Nakashizuka, 1999). Countermeasures might include controlled hunting or fencing, coupled with site preparation and reproduction method cutting across large areas at one time (Sage et al., 2003; Baumhauer et al., 2005).

2.3. Key ecological characteristics of the Fagus species related to regeneration measures

F.s., *F.c.*, *F.g.*, and *F.o.* have shade-tolerant, dominant climax traits that must be considered when planning regeneration projects. Those similarities are central to the “model genus” concept in *Fagus*.

First, several years may pass between masting events, and these cannot be forecast. Second, pollen effectively disperses only about 100 m, or less in closed stands. Third, seeds fall within 20 m of a parent tree. Birds or mammals may carry the nuts farther, but that dispersal is not dependable. Fourth, beech seedlings are sensitive to frost, drought, and animal predation and may suffer from herbaceous competition. Fifth, beech is very shade tolerant and develops best into high quality saplings at sites with dense regeneration and a moderate canopy shelter, at least until the seedlings are large enough for release. Yet those conditions reduce chances for regenerating more light-demanding associated species and increasing tree species diversity.

3. Silvicultural systems and trends in silviculture

Effective silviculture accounts for landowner needs (Nyland, 2002) and prescribes treatments that address specific management goals. As Dengler (1930) stated, silvicultural systems must also take cognizance of ecologic and bio-physical features of a site and the species of interest, i.e. key ecological characteristics.

3.1. Aims in beech management

Except for *F.g.*, beech species are commonly the dominant component of a stand. This often encouraged managers to maintain *F.c.* and *F.s.* in pure stands, using coppice systems for fuelwood production (Kamitani, 1986; Mormiche, 1981 cited in Peters, 1997) and high-forest systems for growing high quality sawtimber. The alternatives of converting beech forests into plantations of faster-growing conifers (e.g., *Cryptomeria* and *Chamaecyparis* in Japan; *Picea* and *Pinus* in Europe) often increased snow and wind breakage, and bark beetle infestations.

Today’s objectives include sustaining multiple services and values from beech forests (Fujimori, 2001; Haynes et al., 2003), often by mimicking the natural dynamics of unmanaged stands, increasing native tree species and their mixtures, and diversifying structures between and within stands (Hahn et al., 2005; Mohadjer, 2005; Wagner and Lundqvist, 2005; Puettmann and Ammer, 2007). Consistent with this, several European countries have major afforestation programmes using beech (e.g., Madsen et al., 2005; Weber, 2005); especially for replacing off-site conifer plantations to restore a more “natural” condition (Spiecker et al., 2004; Bradshaw, 2005; Stanturf, 2005; Hansen and Spiecker, 2005). Further, many researchers propose emulating gap disturbance of unmanaged temperate forests, and that fits the regeneration characteristics of *Fagus*.

Within northeaster North America, beech bark disease makes long-term management of *F.g.* impractical (Nyland et al., 2006b). Landowners can maintain *F.g.* in stands free of beech bark disease or with apparently resistant or tolerant trees (Bogenschütz, 1983), using light partial cutting to promote small beech into overstory positions (Ostrofsky and Houston, 1988). Yet research has not provided effective means for identifying truly resistant clones (Houston, 2003; McCullough et al., 2000), and that complicates management in yet unaffected areas. Among infected stands, understory beech must be controlled in order to successfully regenerate other species (e.g. sugar maple) using shelterwood method and selection system cuttings (Kelty and Nyland, 1981; Ray et al., 1999; Nyland et al., 2006b; Nyland, 2008).

3.2. Regeneration methods

3.2.1. Natural versus artificial regeneration of beech

Managers commonly rely on natural regeneration in beech-dominated forests having an adequate seed source, but may also use direct seeding or planting to introduce new species and provenances. While natural regeneration is considered the least expensive means (Wagner and Lundqvist, 2005), it may fail. In addition, overstory trees left at low density during a long regeneration period may decline in quality and value (Hahn et al., 2005).

Artificial regeneration is appropriate in stands lacking seed trees of desired species or having ones not adapted to a site. In Europe, this is relevant to afforestation programmes on former agricultural land, for advance planting of beech beneath conifer stands, and when a change in beech provenance is required.

3.2.2. Artificial regeneration methods

Artificial regeneration with beech is mostly by planting (e.g., Spiecker et al., 2004; Wagner and Lundqvist, 2005) using bare-root undercut seedlings transplanted from nurseries at 1 to 4 years of age. In Germany, use of wildings has gained popularity (Wagner and Lundqvist, 2005). Container stock seedlings also offer an alternative, including use of 3 to 6 month old stock (Madsen et al., 2006). With beech and oak, container seedlings are more resistant to handling damage and can be used to extend the planting season (Kerr, 1994). Direct seeding is less expensive than these planting methods, and can be used to establish densely stocked regeneration with natural taproots (Bullard et al., 1992; Baumhauer et al., 2005). These grow similar to planted nursery stock (Ammer and Mosandl, 2007). Yet direct seeding requires careful handling of the beechnuts, appropriate seedbed preparation, and careful site selection (Ammer et al., 2002; Löf et al., 2004; Madsen and Löf, 2005). Baumhauer et al. (2005) and Madsen et al. (2006) also recommend using species mixtures and applying the sowing to large areas.

3.2.3. Natural regeneration methods

Taking all ecological characteristics of the genus into account, to succeed with *Fagus,* natural regeneration methods must leave a dense shelter of old trees and require a long regeneration period. With *F.g.*, light partial cutting of single trees or small groups has resulted in an undesirable domination by the beech (Nyland et al., 2006b), i.e. it was very successful for beech regeneration. Similarly, for *F.s.*, those treatments will not ensure a diversity of other species, leading to management problems where the associated ones have important commodity and environmental protection values (Collet et al., 2008).

Shelterwood method with short regeneration periods should ensure a higher species diversity (Kelty and Nyland, 1981; Ray et al., 1999) and result in a cohort density appropriate to high value timber production. However, with *F.o.*, wet autumns, mild winters, cold spring weather, late frosts, frequent droughts, and an acidic humus make success with shelterwood method uncertain (Sagheb-Talebi et al., 2005). With *F.s.*, difficult abiotic conditions, seed-eating animals, and interfering ground vegetation have sometimes caused failures. With *F.c.*, shelterwood method will fail where dwarf bamboo and shrub species interfere with the tree regeneration. Generally, regenerating *F.g.* is not a goal in the aftermath forests of North America (Nyland et al., 2006b).

To this end, there virtually is no single cutting treatment that simultaneously fulfils recent management aims like diversity of species, high quality of young beech, and naturalness with regard to beech dominated natural ecosystems.

4. Some opportunities for regeneration research

No single silvicultural system addresses all ecologic and managerial factors that influence modern beech forestry. Site conditions vary among stands, and different kinds of forest services often depend on maintaining a variety of structural conditions in neighbouring parts of a forest and its adjacent landscape. That may necessitate use of more than one silvicultural system within a forest and across ownerships. To that end, research should continue to explore the responses that follow a wide range of stand treatments, to further articulate the possibilities available to landowners.

Understory root suckers of *F.g.* have reduced shrubs, herbs, and tree regeneration in areas infected with beech bark disease (Forrester and Bohn, 2007), resulting in a gradual increase of beech in aftermath stands. The developing *F.g.* trees die at pole stage, and are replaced by new root suckers, further promoting beech dominance (Nyland et al., 2006a). Research should investigate the resulting stand development pathways and the implications of reduced bio-diversity and simplified structural complexity in affected stands. In addition, field trials should continue to explore non-herbicide methods for controlling *F.g.* to insure favourable species diversity in new cohorts. Managers also need field-expedient methods for identifying tolerant and resistant beech clones to manage for plant species diversity and continued mast production.

Young seedling-origin beech grows more slowly than many associated species (Beaudet and Messier, 1998; Ray et al., 1999; Dreyer et al., 2005), but faster than some. Growth curves for *F.s.* allow managers to compare their development at intermediate ages, but not during early stages of cohort development when cleaning might seem appropriate, as with oaks that need early release to keep them free of oppression by beech (von Lüpke, 1998). This matter requires further attention in research, particularly in regions where global change might alter environmental conditions in ways that affect the relative growth of different species, and where continued beech dominance seems unwise.

Better understanding of the fundamental performance characteristics of trees during the regeneration phase will be needed to insure that silviculture accommodates these potential future conditions. With *F.g.* and *F.s.* the seeds, germinants, and seedlings (Geßler et al., 2007) are drought sensitive, potentially affecting beech seedling regeneration at sites that might be tempered by important climate change. In this regard, Jump et al. (2007) identified temperature as an ecological factor critical to regeneration success at the southernmost distribution of *F.s.*, where it seems limited by moisture stress (Aranda et al., 2000). Similarly, Löf et al. (2005) stressed that drought strongly influences beech performance when light is not limiting, and Bolte et al. (2007) identified a shortage in nitrogen supply among additional drought-induced stresses to beech regeneration. In addition, Lendzion and Leuschner (2008) showed that atmospheric vapour pressure deficits may become limiting to beech regeneration, even with moisture in the rooting medium near optimum. Further, with disturbance-adapted species, irradiance and temperature affect the competitive interference to beech (Fotelli et al., 2004). Thus, gaining greater understanding of the ecophysiological traits of the several beech species and their associates through laboratory research with seedlings has become increasingly important. Regrettably, those investigations with *Fagus* have involved only *F.s.* to date.

Well-documented and widely used beech regeneration strategies of the past may have little value if future weather phenomena differ importantly from those of today's climate. To date, silvicultural research into adapting cutting regimes to accommodate climate change has primarily looked at effects on soil moisture (Czajkowski et al., 2005), including those caused by overstory interception, coupled with soil moisture reduction due to transpiration from the older trees (Wagner et al., 2009). Yet ways to compensate for higher temperatures, reduced precipitation, and vapour pressure deficits remain untested. Research must explore these matters (Lendzion and Leuschner, 2008). Possibilities may include regenerating *F.s.* in mixtures with more drought-tolerant species, and using direct seeding or planting to introduce drought-resistant beech provenances. However, this latter measure may increase risks of introgression to a less well-adapted population, similar to the genetic pollution in wild fruit tree species (for *Malus sylvestris* see Wagner, 2005) and with *Populus nigra* (Heinze, 1998) in Europe. These risks must be evaluated before provenance mixing becomes common with forest tree species like beech.

Despite the historic and continuing northward expansion of *F.c.* on Hokkaido Island (Peters, 1997), ecologists cannot explain the extensive post-glacial dispersal of *F.g.* (Clark et al., 1998), or how some northern outliers of *F.s.* became established in Sweden (Björkman, 1999). Understanding how this dispersion occurred in the past may help forest ecologists to forecast how climate change might alter the future distribution of heavy seeded species like *Fagus*.

Research must also assess new silvicultural options for integrating a broader array of commodity and non-market objectives, and scrutinize recently proposed "near-to-nature" approaches. Studies should differentiate between the nature of a silvicultural system for influencing long-term stand development, and the component treatments that managers might use in at a single point in stand development. Research should also explore ways to coordinate sustained management of single stands with needs and opportunities across an entire forest ownership, at a landscape scale, and across ecological spans of time (Nyland, 2002).

5. Conclusions

This review introduces the idea of a "model-genus" by comparing ecological characteristics of the most prominent *Fagus* species. Findings validate this idea. It enabled us to (i) delineate regeneration strategies that seem appropriate to any of the species of concern, based upon autecological characteristics; (ii) stress the importance of management objectives in silvicultural planning; and (iii) articulate the synecological features critical to successful regeneration. The latter has sometimes been underestimated in the past.

Knowledge derived from this review indicates that management objectives have become increasingly diverse, necessitating a broadening of the silvicultural systems used in these forests. No universal "best-practice" can be applied worldwide, even within the ecological similar genus *Fagus*. This demands continued research into mixing species for high quality timber production, addressing the decline of *Fagus grandifolia* in aftermath forest management, and evaluating *Fagus crenata* management in coexistence with dwarf bamboo. Global climate change presents new challenges, as the long-distance dispersal of heavy seeded *Fagus* species is poorly understood, and the drought sensitive seedlings may succumb to extreme weather.

A model-genus approach should broaden silvicultural knowledge as scientists explore new pathways for managing forests. It may also have relevance with other genera like *Betula* for illustrating the similarities and differences of species with low to intermediate shade tolerance (Perala and Alm, 1990a,b).

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